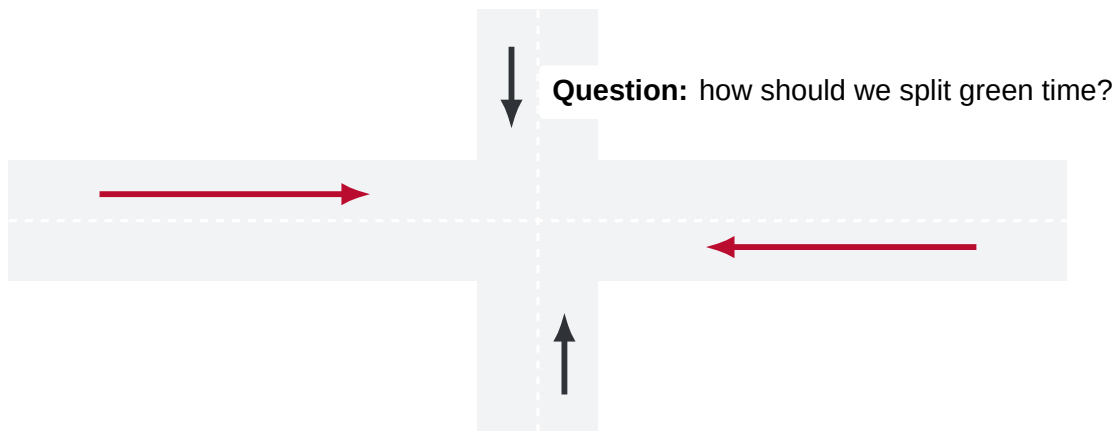


BUCKEYES MATHEMATICAL MODELING AND
SIMULATION SOCIETY

First Meeting Packet

Mathematical Modeling + Traffic Optimization Mini-Project

Today's goal: learn how mathematical modelers turn a messy real-world question into variables, assumptions, equations, computations, and decisions.



Name: _____

Major / Year: _____

Welcome to BMMSS

BMMSS = Buckeyes Mathematical Modeling and Simulation Society.

We are a student community centered on using mathematics, computation, and simulation to understand real systems and make better decisions. The emphasis is not only on getting an answer, but on learning how to *build a defensible model* from an open-ended problem.

What we want members to practice

- Translating real situations into mathematical questions.
 - Choosing variables, parameters, assumptions, and constraints.
 - Building models that are simple enough to analyze but useful enough to inform decisions.
 - Using computation and simulation when exact formulas are not practical.
 - Comparing models with data and revising weak assumptions.
 - Communicating results clearly to technical and non-technical audiences.
- Working in interdisciplinary teams.
 - Seeing connections between mathematics, engineering, computer science, economics, biology, and social science.
 - Developing portfolio-ready modeling projects.
 - Exploring competitions, research directions, and quantitative careers.
 - Learning tools such as Python, spreadsheets, visualization, numerical methods, and LaTeX as projects require them.

You do not need to be an advanced math student to participate. A strong model often begins with careful questions and reasonable assumptions. Different backgrounds contribute different strengths: mathematics, coding, domain knowledge, data analysis, visualization, writing, and presentation.

Suggested flow for today’s meeting

10 min	Introductions and what BMMSS is about
15 min	What mathematical modeling means and where it is used
25 min	Traffic Optimization team mini-project
7 min	Compare models, assumptions, and extensions
3 min	Next steps and future project ideas

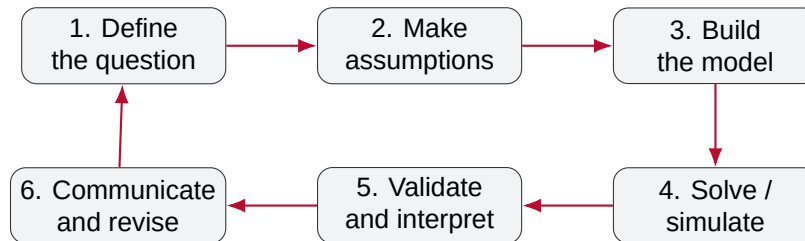
One thing I hope to get from BMMSS:

What Is Mathematical Modeling?

A **mathematical model** is a mathematical representation of a real system. It keeps the features that matter for a particular question and deliberately ignores many details that do not.

Modeling is not “find the formula.” It is a cycle of asking what matters, simplifying reality, building mathematics, checking the consequences, and revising the model.

The modeling cycle



A simple example

Suppose a dining hall wants to reduce lunch lines. “Make the line shorter” is not yet a mathematical problem. A modeler might ask:

- How many students arrive per minute?
- How quickly can each serving station process students?
- What counts as success: average wait, maximum wait, staffing cost, or fairness?
- Does the arrival rate stay constant, or does it spike between classes?

The answers determine the model. Two groups can study the same dining hall and build different valid models because they are answering different questions.

Four pieces you should always identify

Piece	Meaning	Example in traffic
Variables	Quantities we can choose or that change	Green-light time, queue length
Parameters	Inputs treated as known	Arrival rate, road capacity
Constraints	Rules a solution must satisfy	Cycle length, minimum pedestrian time
Objective	What we want to minimize/maximize	Delay, congestion, emissions

Where Is Mathematical Modeling Used?

Area	Example question	Common modeling ideas
Transportation	How should traffic lights be timed?	Optimization, networks, queues, simulation
Public health	How might an epidemic spread?	Differential equations, probability, parameter estimation
Finance	How should risk be allocated across assets?	Probability, statistics, optimization
Engineering	How will heat or stress move through a structure?	Differential equations, numerical methods
Logistics	How should deliveries be routed?	Graphs, integer programming, heuristics
Sports	Which strategy maximizes win probability?	Statistics, stochastic models, game theory
Environment	How do pollutants disperse?	PDEs, simulation, uncertainty analysis
Machine learning	How do we fit a predictive system?	Linear algebra, optimization, probability
Economics	How do incentives change behavior?	Optimization, dynamical systems, game theory
Urban systems	Where should services or resources be located?	Networks, geometry, optimization

What majors belong in modeling?

There is no single “modeling major.” Strong teams are usually interdisciplinary.

Mathematics / Applied Math	Structure, proofs, optimization, differential equations, approximation.
Statistics / Data Science	Estimation, uncertainty, validation, inference, experimental design.
Computer Science	Algorithms, simulation, machine learning, scalable computation.
Engineering	Physical constraints, systems thinking, design tradeoffs, domain models.
Economics / Business	Decision-making, incentives, costs, operations, forecasting.
Natural Sciences	Mechanisms, measurements, conservation laws, domain assumptions.
Social Sciences	Human behavior, survey data, policy effects, interpretation of assumptions.
Communication / Design	Clear visualizations, explanations, presentations, and user-centered decisions.

What mathematics might appear?

Tool	What it helps model
Algebra / functions	Relationships between quantities; simple cost and growth models.
Probability / statistics	Uncertainty, random arrivals, noisy measurements, prediction.
Calculus	Rates of change and continuous optimization.
Linear algebra	Networks, high-dimensional systems, regression, machine learning.
Differential equations	Systems that evolve over time or space.
Discrete mathematics	Routes, schedules, assignments, graphs, combinatorial choices.
Optimization	Choosing the best feasible decision under competing objectives.
Numerical methods	Approximating solutions when exact solutions are unavailable.
Programming	Running experiments, simulations, parameter sweeps, and visualizations.

Key idea: You do not learn every tool first and then begin modeling. Often the project tells you what mathematics you need to learn next.

Mini-Project: Traffic Optimization

Scenario

A city is redesigning the signal timing at a busy four-way intersection. During one representative hour, traffic engineers estimate:

Direction group	Arrival demand	Saturation flow while green
North-South (NS)	700 vehicles/hour	1800 vehicles/hour
East-West (EW)	500 vehicles/hour	1800 vehicles/hour

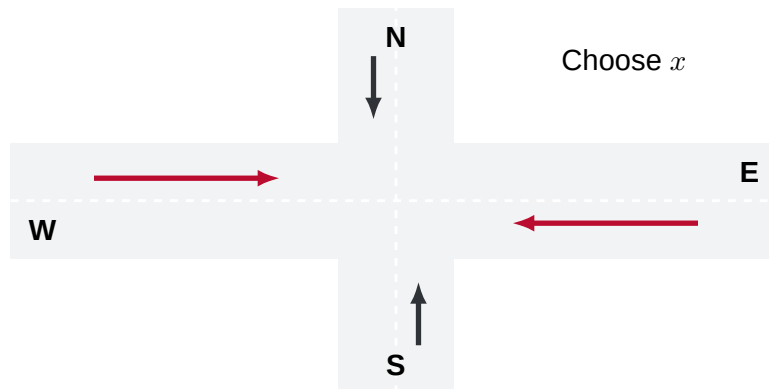
The signal repeats every **90 seconds**. A total of **6 seconds per cycle** is reserved for yellow/all-red transition time, leaving **84 seconds of usable green time**.

Let

$$x = \text{NS green time in seconds per cycle.}$$

Then

$$g_{EW} = 84 - x.$$



Your modeling question: How should the 84 seconds of green time be divided between NS and EW traffic to reduce delay while keeping both directions able to handle demand?

Model assumptions for Round 1

For a first model, assume:

1. Traffic arrives uniformly throughout the cycle.
2. Cars go straight; ignore turning lanes and pedestrians.
3. Each direction can discharge up to 1800 vehicles/hour if it were green continuously.
4. The same signal timing is repeated every cycle.
5. We care about total waiting time, not individual driver fairness yet.

Before calculating: Which assumption seems least realistic, and why?

Traffic Optimization: Team Worksheet

Part A — Feasibility before optimization

If a direction receives a fraction $g/90$ of the cycle as green, approximate its hourly capacity by

$$\text{capacity} = 1800 \left(\frac{g}{90} \right).$$

1. NS capacity. Using $g_{NS} = x$, show that

$$C_{NS}(x) = 20x \quad \text{vehicles/hour.}$$

What must be true about x so NS can handle 700 vehicles/hour?

2. EW capacity. Using $g_{EW} = 84 - x$, write $C_{EW}(x)$ and find the condition on x needed to handle 500 vehicles/hour.

3. Feasible interval. Combine the two conditions:

$$\boxed{\leq x \leq}$$

Part B — Build a waiting-time objective

When traffic arrives uniformly during a red interval of length R , a simple queueing approximation gives total waiting during that red interval as

$$\frac{1}{2} \lambda R^2,$$

where λ is the arrival rate in vehicles/second.

For this intersection:

$$\lambda_{NS} = \frac{700}{3600}, \quad \lambda_{EW} = \frac{500}{3600}.$$

NS is red for $90 - x$ seconds. EW is red for $x + 6$ seconds.

Therefore define total delay per cycle:

$$\boxed{D(x) = \frac{1}{2} \left(\frac{700}{3600} \right) (90 - x)^2 + \frac{1}{2} \left(\frac{500}{3600} \right) (x + 6)^2}$$

measured in **vehicle-seconds per cycle**.

4. Explain the model in words. Why is the heavier NS direction likely to receive more green time?

5. Compare four candidate plans. Complete the table. A calculator is fine.

x (NS green)	EW green	Feasible?	$D(x)$	Avg. delay/vehicle
35	49			
42	42			
50	34			
59	25			

There are about

$$\left(\frac{700 + 500}{3600}\right)(90) = 30$$

vehicles arriving per cycle, so average delay per vehicle is approximately $D(x)/30$.

Traffic Optimization: Find the Best Split

Part C — Optimization

Choose one of the following approaches depending on your background.

Approach 1: Search. Evaluate several feasible integer values of x and identify the smallest delay. You can do this by hand, in a spreadsheet, or with a short program.

Approach 2: Algebra. Expand $D(x)$ into the form $ax^2 + bx + c$ and use the vertex formula

$$x^* = -\frac{b}{2a}.$$

Approach 3: Calculus. Differentiate $D(x)$ and solve $D'(x) = 0$. Then check that the solution is feasible.

6. Your recommended timing:

$g_{NS} =$

sec,

$g_{EW} =$

sec

7. Why is your recommendation better than an equal 42/42 split?

Part D — Critique the model

A useful model is not the same thing as a perfect copy of reality.

8. Pick two missing factors that could materially change the recommendation.

- Left-turning vehicles
 - Pedestrian crossing time
 - Bus stops near the intersection
- Uneven arrival bursts
 - Emergency vehicles
 - Different lane counts

- Driver reaction time
- Construction
- Nearby intersections
- Rain or snow

Factor 1: _____ How would you model it?

Factor 2: _____ How would you model it?

Part E — From one intersection to a city

Suppose the city asks you to coordinate **20 intersections** instead of one. What new mathematical or computational ideas might become useful?

- Graph/network models for the road system.
- Time-dependent traffic flows and queueing models.
- Optimization with hundreds of decision variables and constraints.
- Simulation to test signal plans before deployment.
- Data from sensors, cameras, GPS, or historical traffic counts.
- Multi-objective optimization: delay, pedestrian safety, fuel use, emissions, and fairness.

Modeling lesson: The first model is intentionally simple. Its job is to create a transparent baseline that we understand well enough to criticize and improve.

Discussion: What Did We Actually Model?

In ten minutes of equations, we made many decisions that were not purely mathematical. We chose what “traffic” meant, what counted as “good,” which details mattered, and which facts we were willing to ignore.

Questions for the room

1. Is minimizing total delay always the fairest objective?
2. Should a lightly traveled road ever receive a guaranteed minimum green time?
3. If a pedestrian must wait 90 seconds, should that be treated as equivalent to a driver waiting 90 seconds?
4. How would the optimal timing change during a football game, rush hour, or late at night?
5. If traffic data are uncertain, should we optimize for the average case or protect against bad cases?

Possible future BMMSS project directions

Campus mobility	Shuttle routing, pedestrian flow, parking demand, bike-share balancing.
Sports analytics	Shot selection, player ratings, lineup optimization, win probability.
Public systems	Emergency response location, resource allocation, scheduling.
Networks	Information spread, social networks, disease transmission, internet routing.
Environment	Temperature, pollution, water flow, energy demand.
Machine learning	Prediction compared with interpretable mathematical models.
Operations	Inventory, staffing, queues, scheduling, delivery routes.

A strong BMMSS project should answer four questions:

1. What real decision or phenomenon are we studying?
2. What assumptions make the problem mathematically manageable?
3. What evidence tells us whether the model is useful?
4. What would we improve if we had more time, data, or mathematical tools?

Your idea

A real-world problem I would like BMMSS to model:

Facilitator Reference: Traffic Project

This page can be kept at the back of the packet or separated before distributing the worksheet.

A. Feasible region

NS capacity:

$$C_{NS} = 1800 \frac{x}{90} = 20x.$$

To serve 700 vehicles/hour,

$$20x \geq 700 \implies x \geq 35.$$

EW capacity:

$$C_{EW} = 1800 \frac{84 - x}{90} = 20(84 - x).$$

To serve 500 vehicles/hour,

$$20(84 - x) \geq 500 \implies 84 - x \geq 25 \implies x \leq 59.$$

Thus the feasible interval is

$$\boxed{35 \leq x \leq 59}.$$

B. Candidate plans

x	EW green	Feasible?	$D(x)$ veh-sec/cycle	Avg. sec/vehicle
35	49	Yes	410.83	13.69
42	42	Yes	384.00	12.80
50	34	Yes	373.33	12.44
59	25	Yes	386.82	12.89

C. Optimal split

Differentiate:

$$D'(x) = -\left(\frac{700}{3600}\right)(90 - x) + \left(\frac{500}{3600}\right)(x + 6).$$

Set $D'(x) = 0$:

$$700(90 - x) = 500(x + 6).$$

So

$$63000 - 700x = 500x + 3000,$$

$$60000 = 1200x,$$

$$\boxed{x = 50}.$$

Therefore

$$\boxed{g_{NS} = 50 \text{ sec}, \quad g_{EW} = 34 \text{ sec.}}$$

The second derivative is positive:

$$D''(x) = \frac{700 + 500}{3600} > 0,$$

so this is a minimum, and $x = 50$ lies in the feasible interval $[35, 59]$.

Compared with the equal split, the model reduces delay from 384.00 to 373.33 vehicle-seconds per cycle, about a

$$\frac{384 - 373.33}{384} \times 100\% \approx \boxed{2.8\%}$$

reduction under the stated assumptions.

D. Discussion point

The numerical improvement is modest, which is itself useful. It shows that modelers should ask whether a mathematically optimal change is operationally meaningful, robust to uncertain data, and worth implementing.

Facilitator Run-of-Show

Time	Suggested facilitation
0–5	Welcome. Ask members for name, major, year, and one real-world system they find interesting.
5–10	Explain BMMSS mission: translating open-ended problems into mathematical and computational models.
10–20	Walk through the modeling cycle. Emphasize assumptions, objective, constraints, validation, and revision.
20–25	Show application areas and point out that modeling is interdisciplinary.
25–30	Introduce the traffic scenario. Put members in teams of 3–5.
30–40	Teams complete feasibility and candidate-plan calculations. Circulate and ask why capacity must be checked before optimizing delay.
40–48	Teams find the optimal split using search, algebra, or calculus. Invite different solution methods.
48–55	Compare answers and critique assumptions. Ask which real-world factor they would add first.
55–60	Collect future project ideas and explain how later BMMSS meetings can move from hand models to data, Python, simulation, and larger optimization problems.

Closing message to members: A model is valuable not because it captures everything, but because it makes our assumptions explicit and lets us reason systematically about a real decision. BMMSS is a place to practice that process together.

Notes for next meeting
